Sound Realty ML Model Serving

Technical Architecture & Implementation Deep Dive

ML Engineering Team

2026

Project Overview

- **Goal:** Deploy a KNN model for real-time home price predictions
- **Input Data:** 18 property features + zipcode demographics
- **Model:** K-Nearest Neighbors (k=5) with RobustScaler normalization
- **Stack:** FastAPI, Python, scikit-learn, Docker
- **Architecture:** REST API with hot-reload model capability
- **Outcome:** Production-ready service with comprehensive testing & monitoring

Problem Statement

Challenge: Sound Realty needs to:

- Streamline home valuation process for King County, WA
- Move from manual estimation to automated ML predictions
- Iterate on models without downtime
- Serve predictions at scale with reliability guarantees

Constraints:

- Basic model already exists (numeric features only)
- Need to integrate demographic data from external sources
- Must support A/B testing of different model versions
- Production monitoring & logging required

Data Pipeline & Feature Engineering

Data Sources:

- KC House Sales (21,613 records)
- Zipcode Demographics
- Feature-engineered variants

Feature Set: Primary (bedrooms, bathrooms, sqft_living, grade), Secondary (lat/long, year_built), Demographic (population, income, education)

Data Integration:

```
# Merge demographics by zipcode
merged = sales.merge(demographics, how='left', on='zipcode')
```

Preprocessing: RobustScaler (handles outliers), 80-20 train-test split, Feature importance analysis

Model Architecture: KNN Regression

Why KNN?

- **Interpretability:** Predictions based on k nearest training samples
- **Non-parametric:** Captures complex non-linear relationships
- **No training phase:** Instantaneous retraining possible
- **Baseline:** Good performance on housing datasets

Model Pipeline:

$$\text{RawFeatures} \xrightarrow{\text{RobustScaler}} \text{NormalizedFeatures} \xrightarrow{\text{KNN (k=5)}} \text{Price Prediction}$$

Performance Metrics:

- Test RMSE: \$175K-\$185K (varies by feature set)
- Test MAE: \$105K-\$115K
- Test R^2 Score: 0.82-0.84
- MAPE: 22-24%

Model Versioning & A/B Testing

Multiple Model Variants:

- **Basic Model:** 10 numeric features only
- **Added Features Model:** 18 features + demographics
- **Future Variants:** Gradient Boosting, Neural Networks

Model Storage Structure:

```
model/
            basic/
                  model.pkl (KNN pipeline)
                  model_features.json
                  metrics.json
            added_features/
                  model.pkl
                  model_features.json
            2026-03-05_05/ (timestamp-based)
```

Hot-Reload Capability: Switch models via environment variable without API restart

API Design: REST Endpoints

Endpoint #1: Health Check

```
GET /health      {status: 'healthy', models_loaded: true}
```

Endpoint #2: Full Prediction (18 fields)

```
POST /predict
{
  bedrooms, bathrooms, sqft_living, grade, ..., zipcode
}
    {prediction: $850K, model_version: 'added_features'}
```

Endpoint #3: Minimal Prediction (8 fields)

```
POST /predict-minimal (uses basic model)
```

API Framework: FastAPI

- Async/await for concurrent request handling
- Pydantic validation with automatic OpenAPI docs
- Lifespan context manager for startup/shutdown

Monitoring & Observability

Model Call Logging:

- Every prediction logged with timestamp, input features, output
- Client metadata: IP, user-agent, endpoint
- Latency tracking for performance monitoring
- Structured JSON format for analysis

What We Track:

```
{
  "timestamp": "2026-03-05T10:23:45Z",
  "model_version": "added_features",
  "input_features": {...},
  "prediction": 850000,
  "latency_ms": 12.5,
  "client_ip": "192.168.1.1"
}
```

Benefits:

Testing Strategy

Test Coverage (pytest):

- **API Tests:** Endpoint validation, edge cases, error handling
- **Model Tests:** Prediction outputs, model switching, feature loading
- **Integration Tests:** Full pipeline (data $\rightarrow$ prediction)
- **Logger Tests:** Call logging, JSON serialization

Key Test Scenarios:

- Invalid input handling (missing fields, type mismatches)
- Model hot-reload (switching between versions)
- Prediction consistency (deterministic output)
- Data processing (demographics merge, feature ordering)

CI/CD: pytest automatically runs on every commit

Containerization & Deployment

Docker Strategy:

```
FROM python:3.10-slim
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .
CMD ["uvicorn", "src.api.main:app", "--host", "0.0.0.0"]
```

Docker Compose Orchestration:

- FastAPI service on port 8000
- Nginx reverse proxy on port 80
- Volume mounts for model persistence
- Environment-based model selection

Production Considerations:

- Stateless design for horizontal scaling
- Lazy loading to minimize startup time
- Graceful shutdown with lifespan context

Performance Insights

Model Performance (Test Set):

Metric	Basic	Added Features	Unit
RMSE	180K	175K	$
MAE	110K	105K	$
R^2 Score	0.82	0.84	-
MAPE	23%	22%	%

Bias Analysis:

- Underpricing occurs 18% of the time (bad for sellers)
- Weighted RMSE penalizes underprediction 2x
- Feature importance: sqft_living & grade dominate

Inference Latency: 8-15ms per prediction (GPU not required)

Engineering Decisions & Trade-offs

Decision 1: KNN over Gradient Boosting

- **Pro:** Hot-reload capability, interpretability, fast inference
- **Con:** Slower than GB for very large feature spaces; memory-dependent

Decision 2: RobustScaler over StandardScaler

- **Reasoning:** Housing prices have outliers; RobustScaler is median-based

Decision 3: Lazy Loading of Models

- **Pro:** Fast startup, multiple models in memory
- **Con:** First request slightly slower

Decision 4: Structured Logging

- JSON logs enable machine-readable analysis, drift detection

Future Improvements & Research Directions

Short-term (Next Sprint):

- Ensemble models (voting between KNN, GB, Linear Regression)
- Feature selection optimization (mutual information analysis)
- Hyperparameter tuning (grid search for k value)

Medium-term (Next Quarter):

- Gradient Boosting implementation for ~5% improvement
- Online learning to adapt to market changes
- Geographic clustering (different models per region)

Long-term (Architectural):

- Feature store integration (real-time feature updates)
- Automated model retraining pipeline
- Uncertainty quantification (confidence intervals)
- Multi-objective optimization (minimize RMSE and underpricing)

Production Monitoring Dashboard

Key Metrics to Track:

- **Throughput:** Requests/second
- **Latency:** P50, P95, P99 latencies
- **Model Drift:** Mean absolute error of recent predictions vs. baseline
- **Data Quality:** % of invalid inputs caught by validation
- **Prediction Distribution:** Are we making reasonable price estimates?

Alerting Thresholds:

- API latency ¿ 100ms
- Error rate ¿ 1%
- Prediction mean shifts ¿ 5% from baseline
- Memory usage ¿ 80%

Retraining Trigger: Model drift detected (weekly comparison) or RMSE degrades

Lessons Learned & Best Practices

What Worked Well:

1. Comprehensive logging from day 1 (enabling retrospective analysis)
2. Multiple model versions from the start (no breaking changes)
3. pytest coverage discipline (caught 90% of integration issues)
4. Containerization (reproducible environments)

Technical Best Practices Applied:

1. Separation of concerns (DataLoader, PredictionService, API)
2. Pydantic validation (type safety at boundary)
3. Graceful error handling & structured logging
4. Stateless design (enables horizontal scaling)

Would Do Differently:

- Earlier A/B testing framework (for online metrics)
- Feature monitoring from start (for drift detection)

Questions for Technical Discussion

Potential Discussion Topics:

1. Why not use dimensionality reduction (PCA) given 18 features?
2. How would you handle categorical features (e.g., zipcode as categorical)?
3. What's your strategy for handling missing demographics data?
4. How do you prevent data leakage in train-test split?
5. What about outlier detection before prediction?
6. How would you implement online learning / continual retraining?
7. Prediction intervals vs. point estimates?